Lab Task 08

Network Configuration: Regional Office Connectivity

A software development company has established three regional offices in different cities. Each office has its own local network with multiple computers. The company needs to connect these offices so employees can share resources and communicate across all locations.

You have been hired as a **Network Engineer** to design and implement the inter-office network infrastructure.

Company Requirements:

Office Locations:

1. **Karachi Office (Head Office)**
    - 2 employee computers
    - Local network needs to be established
2. **Lahore Office (Development Center)**
    - 2 developer computers
    - Central location between other offices
3. **Islamabad Office (Sales Office)**
    - 2 sales team computers
    - Need access to resources at both other offices

Business Needs:

- All computers across all three offices must be able to communicate with each other
- Each office should maintain its own separate network

- Network must be reliable and cost-effective
- The Lahore office will serve as the central hub, routing traffic between Karachi and Islamabad

Problem:

Design and configure a network that connects all three regional offices using routers and static routing. Employees from any office should be able to access resources in any other office.

Technical Specifications:

Given Resources:

- 3 Cisco Routers (one per office)
- 3 Switches (one per office)
- 6 Computers (2 per office)
- Copper Cross-Over cables (for router-to-router connections)
- Copper Straight-Through cables (for all other connections)

Network Addressing Scheme:

You must use the following IP addressing plan:

Office	Network Address	Gateway	PC IP Range
Karachi Office	192.168.10.0/24	192.168.10.1	192.168.10.10 to .10.20
Lahore Office	192.168.20.0/24	192.168.20.1	.20.10 to .20.20
Islamabad Office	192.168.30.0/24	192.168.30.1	.30.10 to .30.20
Karachi-Lahore Link	192.168.100.0/24	-	.100.1 and .100.2
Lahore-Islamabad Link	192.168.200.0/24	-	.200.1 and .200.2

Naming Convention:

- Routers: **R_Karachi**, **R_Lahore**, **R_Islamabad**
- Switches: **SW_Karachi**, **SW_Lahore**, **SW_Islamabad**

Your Task:

PART 1: Network Design (2 marks)

Create the network topology in Cisco Packet Tracer showing:

- Proper device placement representing the three offices
- Correct cable connections between all devices
- Labels/notes showing network addresses for each segment

PART 2: Basic Configuration (4 marks)

Configure each network device with:

- Appropriate hostnames
- IP addresses on all router interfaces
- IP addresses, subnet masks, and default gateways on all PCs

PART 3: Static Routing Implementation (3 marks)

Implement static routes so that:

- Karachi office can reach both Lahore and Islamabad offices
- Lahore office can reach both Karachi and Islamabad offices
- Islamabad office can reach both Lahore and Karachi offices
- **All routes must be explicitly configured - no default routes allowed**

PART 4: Testing & Verification (2 marks)

Demonstrate successful configuration by:

- Showing routing tables from all three routers
- Testing connectivity between different offices (minimum 4 different tests)
- Providing a traceroute from Karachi to Islamabad showing the complete path
- Explaining why Lahore router requires different routing logic than the other two

Submission Instructions:

Two files are required for submission:

- **Cisco Packet Tracer File (.pkt)**:
    - File name format: Section_RollNumber.pkt
    - Must contain complete working topology with all configurations saved
- **Documentation Report (Word or PDF)**:
    - File name format: Section_RollNumber.docx or .pdf
    - **Must include:** Network topology diagram, IP addressing table, router configurations, routing tables (show ip route screenshots), connectivity testing (minimum 4 ping tests), traceroute output, and detailed answers to all analysis questions

ALL screenshots in your report must display your Roll Number using the Note tool in Cisco Packet Tracer. Screenshots without roll numbers will not be accepted